

COMP8851 - Master Results Table

Group: MDS-IoT-Yu2 | Supervisor: Dr Yu Zhang

How to use this document

This table contains all model results collected from every team member's improvement report. Use these numbers to complete your Reflection Part 2 (Cross-Model) section. Green highlighted rows indicate the best performing result in each metric column. When writing your reflection refer to specific numbers from this table rather than making general claims.

Table 1 - Overall MAE and RMSE (All Models)

Model	Contributor	MAE (degrees)	RMSE (degrees)	Training Time
EKF Baseline	Md Shahrear Bhuiyan	86.79	102.13	N/A
MLP Baseline	Anuj Adhikari	68.75	85.37	2.79 min (GPU)
TCN Baseline	Sarum Younas	69.27	85.99	620.94 min (MPS)
LSTM Initial	Sai Manoj Polina	67.05	83.61	345.76 min (CPU)
LSTM Refined	Sai Manoj Polina	69.22	86.08	620.36 min (CPU)
Transformer Initial	Ankita Deepak Dimri	66.99	83.30	~80 min (GPU)
Transformer Refined	Ankita Deepak Dimri	66.65	82.44	~80 min (GPU)
Hybrid Initial	Gopal Datt Bhatt	66.31	82.99	387.06 min (CPU)
Hybrid Refined	Gopal Datt Bhatt	66.77	83.53	~387 min (CPU)

Note: Transformer Refined achieved the best MAE among improved models (66.65 degrees). Hybrid Initial achieved the best overall MAE across all configurations (66.31 degrees). LSTM Refined performed worse than the original, confirming that increased dropout and wider windows hurt overall generalisation for LSTM.

Table 2 - Scenario Breakdown MAE (degrees)

Model	Straight Walking	Sharp Turns	Short Trajectories	Long Trajectories
EKF Baseline	N/A	N/A	N/A	N/A
MLP Baseline	62.63	88.73	65.62	70.80
TCN Baseline	63.39	87.88	74.38	64.15
LSTM Initial	59.96	86.07	69.03	65.08
LSTM Refined	61.00	84.44	73.43	65.01
Transformer Initial	60.57	86.05	68.09	66.08
Transformer Refined	59.06	82.19	69.14	64.17
Hybrid Initial	59.26	89.66	65.06	67.14
Hybrid Refined	60.04	88.54	63.67	68.83

Note: Transformer Refined achieved the best sharp turn MAE of 82.19 degrees, the most significant scenario-level improvement across all models. Hybrid Initial has the worst sharp turn MAE at 89.66 degrees despite having the best overall MAE. This contradiction is the key analytical finding for the final report discussion.

Table 3 - Final Ranking by Overall MAE (Best to Worst)

Rank	Model	MAE (degrees)	RMSE (degrees)
1	Hybrid Initial	66.31	82.99
2	Transformer Refined	66.65	82.44
3	Transformer Initial	66.99	83.30
4	LSTM Initial	67.05	83.61
5	Hybrid Refined	66.77	83.53
6	MLP Baseline	68.75	85.37
7	TCN Baseline	69.27	85.99
8	LSTM Refined	69.22	86.08
9	EKF Baseline	86.79	102.13

Key Findings for Phase 2 Reflections

1. All learned models significantly outperform EKF, confirming deep learning is superior to filter-based heading estimation.
2. MLP at 68.75 degrees outperforms TCN (69.27 degrees) and improved LSTM (69.22 degrees), showing that adding temporal structure does not always guarantee improvement.
3. Transformer Refined achieved the best sharp turn MAE of 82.19 degrees, a 3.86 degree improvement over the original, driven by the wider 400-timestep window.
4. Hybrid Initial has the best overall MAE (66.31 degrees) but worst sharp turn MAE (89.66 degrees) among deep learning models. This contradiction is the central analytical finding.
5. The gap between all deep learning models is very narrow (within 2.91 degrees MAE), suggesting subject generalisation is the primary bottleneck rather than architecture choice.
6. LSTM Refined performed worse than the original despite targeted regularisation, confirming LSTM architecture limitations rather than training setup as the bottleneck.